

Online Appendix

Consistency across WTP Elicitation Formats under Valuation Uncertainty: Evidence from a
Large-Scale Water Diversion Project in China

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A Descriptive Statistics

Table A1: Some categorical descriptive statistics (1 of 3)

Variables	DBDC	MBDC
Respondent's characteristics		
Age		
0-24 years old	0.071	0.062
25-39 years old	0.268	0.271
40-65 years old	0.507	0.492
Above 65 years old	0.154	0.175
Household characteristics		
Family monthly income per capita		
0-500	0.040	0.039
500-1,000	0.107	0.135
1,000-2,000	0.296	0.271
2,000-4,000	0.398	0.372
4,000-6,000	0.107	0.124
Above 6,000	0.073	0.059

Table A2: Some categorical descriptive statistics (2 of 3)

Variables	DBDC	MBDC
Water scarcity		
Score of tap water pressure (1 to 5: bad to good)		
Score 1	0.057	0.047
Score 2	0.481	0.468
Score 3	0.356	0.364
Score 4	0.100	0.105
Score 5	0.007	0.016
Frequency of water outage in the past year		
Freq. 0	0.533	0.524
Freq. 1-2	0.393	0.368
Freq. above 2	0.074	0.108
Longest outage duration in the past year (hour)		
Less than 1	0.528	0.538
1-2	0.381	0.391
Above 3	0.091	0.071

Table A3: Some categorical descriptive statistics (3 of 3)

Variables	DBDC	MBDC
Water quality		
Score of tap water turbidity (1 to 5: bad to good)		
Score 1	0.038	0.031
Score 2	0.175	0.168
Score 3	0.353	0.337
Score 4	0.396	0.402
Score 5	0.038	0.062
Score of tap water impurity (1 to 5: bad to good)		
Score 1	0.031	0.029
Score 2	0.199	0.201
Score 3	0.301	0.294
Score 4	0.438	0.425
Score 5	0.031	0.051
Score of overall tap water quality (1 to 10: bad to good)		
Score 1-2	0.178	0.036
Score 3-4	0.397	0.081
Score 5-6	0.239	0.392
Score 7-8	0.099	0.358
Score 9-10	0.087	0.133

Table A4: Distribution of DBDC Responses

Bids	YY (%)	YN (%)	NY (%)	NN (%)
2-4-1	80.5	10.4	2.7	5.5
4-8-2	60.2	19.3	8.6	10.7
8-15-4	50.3	22.6	9.9	16.3
15-30-8	25.1	28.7	20.8	23.0

Note: YY, YN, NY, and NN denote response options in the DBDC format, where Y and N indicate yes and no responses to sequential bid offers.

Table A5: Distribution of MBDC Responses

Bids (CNY)	Definitely yes (%)	Probably yes (%)	Not sure (%)	Probably not (%)	Definitely not (%)
0	100.0	0.0	0.0	0.0	0.0
0.5	87.4	7.8	0.0	0.2	4.5
1.0	82.5	10.7	2.1	0.0	4.7
1.5	77.3	11.4	4.5	1.7	5.2
2.0	72.7	14.0	4.5	2.4	6.4
3.0	62.8	16.6	8.8	3.8	8.1
4.0	57.3	16.4	9.5	7.3	9.5
5.0	54.0	14.5	10.9	7.8	12.8
6.0	43.4	14.5	14.0	8.1	20.1
7.0	42.2	12.6	11.6	9.7	23.9
8.0	40.8	10.7	11.4	9.0	28.2
10	37.4	11.6	7.8	9.0	34.1
15	18.7	20.4	10.9	9.0	41.0
20	15.9	12.6	12.8	11.4	47.4
27	8.8	12.1	10.2	13.7	55.2
35	6.6	9.2	11.4	10.4	62.3
42	5.0	6.2	8.5	12.1	68.2
50	4.7	5.0	6.6	8.8	74.9
65	2.1	4.0	3.8	8.8	81.3
80	1.9	1.9	4.3	6.2	85.8
100	1.2	2.1	2.6	4.0	90.0
150	0.7	1.4	1.7	2.6	93.6
200	0.7	1.2	0.5	2.1	95.5
250	0.7	0.9	0.5	0.5	97.4
300	0.7	0.9	0.2	0.0	98.1

Note: This table reports the percentage distribution of respondents across five uncertainty categories at each bid amount under MBDC format.

B Principle Component Analysis

Table B1: PCA for Water Scarcity Index

Variable	DBDC			MDBC		
	PC1	PC2	PC3	PC1	PC2	PC3
Water pressure	-0.277	-0.939	0.202	-0.305	-0.826	0.193
Outage frequency	0.698	0.052	-0.714	0.638	0.105	-0.628
Outage duration	0.660	0.339	0.671	0.625	0.407	0.618
Eigenvalue	1.577	0.976	0.447	1.617	0.925	0.459
Proportion explained	0.526	0.325	0.149	0.539	0.308	0.153
Cumulative explained	0.526	0.851	1.000	0.539	0.847	1.000

Note: This table reports PCA results used to construct the water scarcity index. The first principal component is used as the water scarcity index in the main analysis.

Table B2: PCA for Water Quality Index

Variable	DBDC			MDBC		
	PC1	PC2	PC3	PC1	PC2	PC3
Water turbidity	0.608	-0.294	-0.736	0.539	-0.286	-0.762
Water impurity	0.594	-0.445	0.669	0.462	-0.436	0.681
Water quality score	0.525	0.845	0.096	0.582	0.804	0.105
Eigenvalue	1.865	0.673	0.461	2.145	0.546	0.309
Proportion explained	0.621	0.224	0.153	0.715	0.182	0.103
Cumulative explained	0.621	0.846	1.000	0.715	0.897	1.000

Note: This table reports PCA results used to construct the water quality index. The first principal component is used as the water quality index in the main analysis.

Table B3: PCA for Water Knowledge

Variable	DBDC				MDBC			
	PC1	PC2	PC3	PC4	PC1	PC2	PC3	PC4
Number of known water sources	0.629	-0.261	-0.137	-0.718	0.682	-0.243	-0.167	-0.709
Knows main water source	0.601	-0.405	-0.059	0.685	0.617	-0.417	-0.082	0.638
Receives water-related news	0.315	0.746	-0.574	0.114	0.325	0.738	-0.561	0.138
Knows water price	0.377	0.458	0.804	0.010	0.338	0.462	0.828	0.017
Eigenvalue	1.588	0.977	0.877	0.556	1.312	1.180	0.952	0.556
Proportion explained	0.397	0.244	0.219	0.139	0.328	0.295	0.238	0.139
Cumulative explained	0.397	0.641	0.861	1.000	0.328	0.623	0.861	1.000

Note: This table reports PCA results used to construct the water knowledge index. The first principal component is used as the water knowledge index in the main analysis.

C Wang-type First Stage

Table C1: MBDC Estimates with Wang-type First Stage

	Mean	Std. err.	Min.	Max.
	(1)	(2)	(3)	(4)
Estimated WTP (μ_i)	22.54	20.41	0.06	233.10
Estimated WTP std. err. (σ_i)	9.84	14.96	0.02	177.31
Observation	422			

Note: This table reports first-stage estimates of each individual's WTP under the Wang-type MBDC specification. The individual-level mean and standard deviation of WTP are obtained from equation (3). We recover the WTP instead of presenting the log form.

D Normal Distribution

Table D1: DBDC Estimates

	Coef. (1)	Std. err. (2)
Respondent's characteristics		
Age	-0.026	(0.007)
Female	-0.238	(0.046)
College degree	0.219	(0.069)
Household characteristics		
Monthly income per capita (log)	0.027	(0.013)
Monthly water bill	0.003	(0.005)
Installation of water filters	0.034	(0.167)
Water usage		
Water scarcity index	0.111	(0.036)
Water quality index	-0.438	(0.121)
Water knowledge index	0.197	(0.064)
Other parameter		
Intercept	2.918	(0.847)
Observation	610	
Log likelihood	-682.598	

Note: This table reports DBDC estimates of households' WTP determinants assuming a normal distribution. Coefficients are obtained from equation (1). Standard errors are in parentheses.

Table D2: MBDC Estimates with Welsh-type

	Definitely yes (1)	Probably yes (2)	Not sure (3)	Probably no (4)	Definitely no (5)
Respondent's characteristics					
Age	-0.014 (0.005)	-0.012 (0.004)	-0.017 (0.005)	-0.013 (0.004)	-0.016 (0.005)
Female	-0.154 (0.045)	-0.167 (0.041)	-0.160 (0.042)	-0.158 (0.043)	-0.169 (0.041)
College degree	0.298 (0.098)	0.322 (0.107)	0.339 (0.113)	0.347 (0.104)	0.336 (0.111)
Household characteristics					
Monthly income per capita (log)	0.020 (0.007)	0.022 (0.006)	0.023 (0.011)	0.020 (0.007)	0.022 (0.010)
Monthly water bill	0.003 (0.004)	0.002 (0.003)	0.001 (0.003)	0.001 (0.003)	0.002 (0.004)
Installation of water filters	0.062 (0.121)	0.071 (0.122)	0.059 (0.124)	0.073 (0.120)	0.058 (0.123)
Water usage					
Water scarcity index	0.168 (0.079)	0.149 (0.078)	0.159 (0.077)	0.155 (0.078)	0.164 (0.079)
Water quality index	-0.432 (0.081)	-0.524 (0.086)	-0.471 (0.082)	-0.487 (0.083)	-0.495 (0.082)
Water knowledge index	0.112 (0.023)	0.133 (0.027)	0.145 (0.022)	0.119 (0.019)	0.126 (0.026)
Other parameters					
Intercept	2.201 (0.471)	2.598 (0.613)	3.118 (0.732)	3.402 (0.744)	3.742 (0.821)
Std. err. ε_i	0.648 (0.209)	0.739 (0.214)	0.921 (0.295)	1.048 (0.337)	1.152 (0.338)
Observation	422				
Log likelihood	-1315.284	-1314.267	-1368.249	-1312.214	-1311.284

Note: This table reports Welsh-type MBDC estimates of households' WTP determinants assuming normal distribution. Coefficients are obtained from equation (2). Standard errors are in parentheses.

Table D3: MBDC Estimates with Wang-type First Stage

	Mean (1)	Std. err. (2)	Min. (3)	Max. (4)
Estimated WTP (μ_i)	22.50	30.46	0.07	233.14
Estimated WTP std. err. (σ_i)	9.88	14.90	0.03	177.28
Observation	422			

Note: This table reports first-stage estimates of each individual's WTP under the Wang-type MBDC specification with the normal distribution. The individual-level mean and standard deviation of WTP are obtained from equation (3). We recover the WTP instead of presenting the log form.

Table D4: MBDC Estimates with Wang-type Second Stage

	Coef. (1)	Std. err. (2)
Respondent's characteristics		
Age	-0.024	(0.006)
Female	-0.221	(0.072)
College degree	0.233	(0.067)
Household characteristics		
Monthly income per capita (log)	0.027	(0.014)
Monthly water bill	0.003	(0.005)
Installation of water filters	0.031	(0.168)
Water usage		
Water scarcity index	0.109	(0.032)
Water quality index	-0.417	(0.118)
Water knowledge index	0.203	(0.064)
Other parameter		
Intercept	2.764	(0.812)
Observation	422	
R-squared	0.829	

Note: This table reports Wang-type MBDC second-stage estimates of households' WTP determinants assuming normal distribution. Coefficients are obtained from equation (4). Standard errors are in parentheses.

E Alternative Probability Assignments

Table E1: MBDC Estimates with Wang-type First Stage

	Mean (1)	Std. err. (2)	Min. (3)	Max. (4)
Estimated WTP (μ_i)	22.54	30.50	0.06	233.18
Estimated WTP std. err. (σ_i)	9.93	14.94	0.04	177.33
Observation	422			

Note: This table reports first-stage estimates of each individual's WTP under the Wang-type MBDC specification with probabilities of 0.9, 0.7, 0.5, 0.3, and 0.1. The individual-level mean and standard deviation of WTP are obtained from equation (3). We recover the WTP instead of presenting the log form.

Table E2: MBDC Estimates with Wang-type Second Stage

	Coef. (1)	Std. err. (2)
Respondent's characteristics		
Age	-0.026	(0.007)
Female	-0.242	(0.074)
College degree	0.219	(0.064)
Household characteristics		
Monthly income per capita (log)	0.029	(0.014)
Monthly water bill	0.003	(0.005)
Installation of water filters	0.033	(0.173)
Water usage		
Water scarcity index	0.111	(0.033)
Water quality index	-0.439	(0.121)
Water knowledge index	0.214	(0.068)
Other parameter		
Intercept	2.903	(0.847)
Observation	422	
R-squared	0.818	

Note: This table reports Wang-type MBDC second-stage estimates of households' WTP determinants with probabilities of 0.9, 0.7, 0.5, 0.3, and 0.1. Coefficients are obtained from equation (4). Standard errors are in parentheses.

Table E3: MBDC Estimates with Wang-type First Stage

	Mean (1)	Std. err. (2)	Min. (3)	Max. (4)
Estimated WTP (μ_i)	22.48	30.43	0.08	233.11
Estimated WTP std. err. (σ_i)	9.86	14.93	0.05	177.26
Observation	422			

Note: This table reports first-stage estimates of each individual's WTP under the Wang-type MBDC specification with probabilities of 0.95, 0.7, 0.5, 0.3, and 0.05. The individual-level mean and standard deviation of WTP are obtained from equation (3). We recover the WTP instead of presenting the log form.

Table E4: MBDC Estimates with Wang-type Second Stage

	Coef. (1)	Std. err. (2)
Respondent's characteristics		
Age	-0.027	(0.008)
Female	-0.229	(0.071)
College degree	0.241	(0.069)
Household characteristics		
Monthly income per capita (log)	0.026	(0.013)
Monthly water bill	0.003	(0.004)
Installation of water filters	0.029	(0.169)
Water usage		
Water scarcity index	0.113	(0.035)
Water quality index	-0.411	(0.115)
Water knowledge index	0.198	(0.063)
Other parameter		
Intercept	2.912	(0.831)
Observation	422	
R-squared	0.804	

Note: This table reports Wang-type MBDC second-stage estimates of households' WTP determinants with probabilities of 0.95, 0.7, 0.5, 0.3, and 0.05. Coefficients are obtained from equation (4). Standard errors are in parentheses.

F Oath Script

Table F1: MBDC Estimates and Oath Scripts

	Coef. (1)	Std. err. (2)
Oath scripts		
Oath scripts	-0.128	(0.028)
Respondent's characteristics		
Age	-0.024	(0.006)
Female	-0.221	(0.044)
College degree	0.231	(0.073)
Household characteristics		
Monthly income per capita (log)	0.029	(0.014)
Monthly water bill	0.005	(0.005)
Installation of water filters	0.031	(0.158)
Water usage		
Water scarcity index	0.109	(0.034)
Water quality index	-0.417	(0.119)
Water knowledge index	0.203	(0.067)
Other parameter		
Intercept	2.764	(0.812)
Observation	511	
R-squared	0.863	

Note: This table reports Wang-type MBDC second-stage estimates of households' WTP determinants with the inclusion of oath scripts. Coefficients are obtained from equation (4). Standard errors are in parentheses.

G Statistical Efficiency of Estimates

As the final component of the comparison, we evaluate the relative statistical efficiency of the three estimation models using the trace (A-optimality) and the determinant (D-optimality) criteria (Dette, 1997; Jones et al., 2021). Because the parameter vector β exists in all specifications, the comparison is based on the estimated covariance matrix of $\hat{\beta}$. Specifically, we consider the following two statistics.

$$\text{tr}(\hat{\Sigma}_\beta) \quad \text{and} \quad \det(\hat{\Sigma}_\beta)^{1/k_\beta} \quad (1)$$

where $\text{tr}(\cdot)$ denotes the trace, $\det(\cdot)$ denotes the determinant, and k_β is the dimension of the parameter vector β . The matrix $\hat{\Sigma}_\beta$ is defined as:

$$\hat{\Sigma}_\beta = n \cdot \widehat{\text{Var}}(\hat{\beta}), \quad (2)$$

which scales the estimated covariance matrix by the sample size for consistent comparisons across estimation models. Smaller values of either criterion indicate greater statistical efficiency.

The covariance matrix $\hat{\Sigma}_\beta$ is derived from the Fisher information matrix according to the DBDC and Welsh MBDC specifications by a one-step maximum likelihood estimation. In contrast, under the Wang-type MBDC approach, which follows a two-step estimation procedure, $\hat{\Sigma}_\beta$ is obtained from the corresponding Godambe information and takes a sandwich form. To ensure comparability between models and to accommodate potential deviations from parametric assumptions, we compute $\hat{\Sigma}_\beta$ using a bootstrap procedure for all three specifications.

We show the comparison of statistical efficiency in Appendix Table G1. The Welsh-type MBDC specification is the most efficient because it uses multiple valuation cutoffs per respondent and estimates all parameters jointly in a single-step maximum likelihood framework. The DBDC approach is less efficient than the Welsh-type MBDC because each respondent provides only two valuation points, but it remains relatively efficient due to its single-step estimation and direct use of the Fisher information matrix. In contrast, the Wang-type MBDC approach exhibits lower efficiency because it relies on a two-step procedure with generated regressors, and inference is based on the Godambe and sandwich form rather than the complete Fisher information matrix, resulting in larger estimated variances for β .

Table G1: Relative Statistical Efficiency Compared to DBDC

	$\text{tr}(\widehat{\Sigma}_\beta)$	$\det(\widehat{\Sigma}_\beta)^{1/k_\beta}$
Welsh-type MBDC		
Definitely yes	0.62	0.70
Probably yes	0.65	0.72
Not sure	0.64	0.77
Probably no	0.60	0.71
Definitely no	0.67	0.79
Wang-type MBDC	1.21	1.27

Note: This table reports relative statistical efficiency measures for the Welsh-type and Wang-type MBDC estimates, normalized to the DBDC benchmark. The trace and determinant of the estimated covariance matrix for the DBDC specification are normalized to one. Lower values indicate higher statistical efficiency relative to the DBDC baseline.

References

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